



Module 2 challenge

✓ 1. What is the difference between stream ingestion and batch ingestion?

- ☐ Both stream and batch ingestion process data in predefined intervals or larger chunks, but they transform the data in different ways.
- ☐ Both stream and batch ingestion involve real-time continuous processing of data, but they process different types of data.
- ✓ ☒ Stream ingestion involves real-time continuous processing of data, while batch ingestion processes data in predefined intervals or larger chunks.
- ☐ Stream ingestion processes data in predefined intervals or larger chunks, while batch ingestion involves real-time continuous processing of data.

That is correct!

✓ 2. You need to process a large amount of raw data and change it into a format for easy analysis. This process will affect your company's storage costs in several ways. Which factors can significantly affect storage costs?

- ✓ ☒ Faster data processing requires more advanced storage solutions.
- ☐ More data processing results in less storage space needed.
- ☐ More data processing results in cheaper storage solutions.
- ☐ Slower data processing requires more advanced storage solutions.

That is correct!

✓ 3. A university wants to optimize its course offerings based on student preferences and performance data. Which benefit of cloud data analytics would be most helpful for the university?

✓ Receiving insight of the evaluation and effectiveness of courses and content

☐ Making it easier to transfer budgetary resources between departments

☐ Making it easier for students to transfer between academic institutions

☐ Receiving faculty feedback early and often to improve the teaching experience

That is correct!

✗ 4. An organization decides to implement cloud cost optimization. What is the primary goal of cloud cost optimization for the organization?

☐ To decrease carbon emissions and improve infrastructure effectiveness

✗ ~~To decrease cloud expenses and increase data effectiveness~~

☐ To enhance the performance of applications and reduce workload

☐ To enhance workload efficiency and get maximum value from cloud resources

Review the video about cost optimization for the cloud.

Ans- To enhance workload efficiency and get maximum value from cloud resources

- ✗ 5. In cloud architecture, backend platforms are responsible for data storage. What two additional roles do backend platforms have in cloud architecture? Select two answers.

ⓘ Note: To get credit for a multiple-select question, you must select all of the correct options and none of the incorrect ones.

✗ ~~Application performance~~

Review the video overview of cloud architecture.

☐ Security mechanisms

✓ Management

That is correct!

☐ Data deletion

☐ Cost visibility

Ans- security mechanism

- ✗ 6. You want to store a piece of information with a unique identifier which can be found no matter where it is stored. Which type of data should you use?

✗ ~~Block data~~

- ☐ File data
- ☐ Object data
- ☐ Stream data

Review the video about the cloud's role in advancing the field of analytics.

Ans- Object data

- ✓ 7. Your company wants to purchase resources from a cloud provider through a website or online portal and have the resources quickly made available, usually within hours or even minutes. Which resource provisioning model would you recommend?

- ☐ Dynamic provisioning
- ☐ Advanced provisioning
- ☐ Fixed provisioning
- ✓ Self-provisioning

That is correct!

✓ 8. What data pricing model is based on the computing power used to run queries over time and measured in virtual central processing units, called slots?

- ☐ Dynamic provisioning
- ☐ On-demand
- ☐ Advanced provisioning
- ✓ ☒ Capacity

That is correct!

✓ 9. An organization wants to guarantee that its applications run smoothly while reducing cloud expenses. What key benefit of cloud cost optimization can help them achieve this goal?

- ☐ Rightsizing
- ☐ Auto-scaling
- ✓ ☒ Improved performance of applications
- ☐ Reduced cost visibility

That is correct!

✗ 10. You are planning to use a cloud service to hold and organize a large amount of data. Which component of cloud architecture will be most involved in this task?

☐ Storage

☐ Application

☐ Service

✗ ~~Runtime cloud~~

Review the video about overview of cloud architecture.

Ans- storage

✓ 1. A company wants to transform its data after it has been loaded into the warehouse. Which process should the company use?

✓ ☒ Extract, Load, Transform (ELT)

☐ Transform, Extract, Load (TEL)

☐ Load, Extract, and Transform (LET)

☐ Extract, Transform, and Load (ETL)

That is correct!

✓ 2. What does the term “network use” refer to in the context of cloud expenses?

☐ The computing power used to run queries over time

☐ The amount of data processed during query processes

✓ ☒ The amount of data that is read or moved between storage buckets

☐ The amount of data kept in storage

That is correct!

✓ 3. A manufacturing company is facing challenges like unpredictable demand fluctuations and supply chain disruptions. Which benefit of cloud data analytics would be most helpful for this company?

- ☐ Making it easier for customers to describe project requirements
- ☐ Improving the consistency of data from the supply chain
- ☐ Evaluating customer feedback of the manufacturer's products
- ✓ ☒ Making sure manufacturers produce exactly what their customers want

That is correct!

✓ 5. You are shopping online. You add items to your cart and proceed to checkout. What backend component in cloud architecture enables this experience?

- ☐ Service
- ✓ ☒ Application
- ☐ Runtime cloud
- ☐ Infrastructure

That is correct!

- ✓ 6. The IT team for a tech company is using several types of tools to support its internal operations. The IT team uses web interfaces, application programming interfaces (APIs), Pub/Sub, Looker, and Jupyter Notebooks. What data tasks would the IT team use these tools to conduct?

- ☐ To adjust computing resources
- ✓ ☒ To access data stored in the cloud
- ☐ To build on-site storage solutions
- ☐ To streamline supply chain systems

That is correct!

- ✓ 8. How does integrating cloud data analytics into manufacturing processes help companies?

- ☐ By reducing the amount of data retrieved from manufacturing processes
- ☐ By increasing the quality of manufactured products
- ☐ By sharing supply chain disruptions with customers
- ✓ ☒ By optimizing operations for reliability and efficiency

That is correct!

✓ 10. Based on how it operates, what computing concept is runtime cloud similar to?

☐ Data storage

☐ A user interface

✓ ☒ An operating system

☐ Frontend platforms

That is correct!